Overall Survival Prediction in Adult Myeloid Leukemia: Kaplan–Meier, Cox Regression and Machine-Learning Survival Models

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Abstract

This report documents my work on the QRT × Institut Gustave Roussy data challenge on Challenge Data (ENS Paris), whose goal is to rank adult patients with myeloid leukemia by their risk of death, using clinical measurements, cytogenetic reports and somatic gene mutations. Models are scored with the inverse probability of censoring weighted (IPCW) concordance index truncated at seven years. This draft covers the data audit. Its main finding is that the 1,193 test patients all come from a single clinical centre that is absent from the 23 training centres, and that this centre's patients are clinically sicker, with more than twice the training rate of *TP53* mutations. Model validation will therefore hold out whole centres.

1 Introduction

Myeloid leukemias are blood cancers in which abnormal immature cells (blasts) accumulate in the bone marrow. Estimating a patient's prognosis guides treatment: low-risk patients may receive supportive care, while high-risk patients may be prioritised for intensive treatment such as stem cell transplantation.

The outcome is *overall survival* (OS): the time from diagnosis to death or to the last follow-up. When a patient is still alive at the last follow-up, the true survival time is unknown; we only know it is longer than the observed time. Such observations are called *right-censored*, and handling them correctly is the purpose of survival analysis.

The project follows a staged plan: Kaplan–Meier estimation and log-rank tests, Cox proportional hazards regression, and machine-learning survival models (gradient-boosted survival and random survival forests), all compared under the challenge metric with honest cross-validation. A similar crowdsourced effort, the DREAM 9 AML Outcome Prediction Challenge, also scored survival models with a concordance index [1].

2 Data

The data come from 24 clinical centres and are split into the five files described in Table 1. The `ID` column links the files.

Table 1: The five challenge files.

File	Rows	Content
`clinical_train.csv`	3,323	one row per patient: centre, blood counts, karyotype
`clinical_test.csv`	1,193	same, for test patients
`molecular_train.csv`	10,935	one row per somatic mutation
`molecular_test.csv`	3,089	same, for test patients
`target_train.csv`	3,323	`OS_YEARS`, `OS_STATUS`

Clinical variables. `BM_BLAST` (bone-marrow blasts, %), `WBC` (white blood cells, G/L), `ANC` (absolute neutrophil count, G/L), `MONOCYTES` (G/L), `HB` (haemoglobin, g/dL), `PLT` (platelets, G/L) and `CYTOGENETICS`, a free-text karyotype written in ISCN notation (for example `46,xy` for a normal male karyotype).

Molecular variables. Each row is one somatic mutation, with its chromosomal position, reference and alternate bases, the gene, the protein change, the broad effect, the variant allele fraction (`VAF`, the share of sequencing reads carrying the mutation) and, not mentioned in the challenge description, the sequencing read depth (`DEPTH`).

3 The evaluation metric

For each patient i we observe a time T_i (`OS_YEARS`), an event indicator Δ_i (1 = death, 0 = censored), and the model gives a risk score η_i.

Harrell's C-index. A pair of patients (i, j) is *comparable* when we know who died first, that is, when $T_i < T_j$ and patient i died ($\Delta_i = 1$). The pair is *concordant* when the model agrees, $\eta_i > \eta_j$. Harrell's C-index [2] is

$$C = \frac{\text{number of concordant comparable pairs}}{\text{number of comparable pairs}}. \tag{1}$$

A value of 0.5 corresponds to random guessing and 1 to a perfect ranking.

IPCW C-index (Uno's C). Patients who die late are less likely to be observed dying, because many patients are censored before then. Harrell's C therefore over-represents early deaths. Uno *et al.* [3] correct this by weighting each comparable pair by $1/\hat{G}(T_i)^2$, where $\hat{G}(t)$ is the Kaplan–Meier estimate of the probability of *not yet being censored* at time t. With the truncation horizon $\tau = 7$ years used by the challenge,

$$C_\tau = \frac{\displaystyle\sum_i \sum_j \Delta_i \, \hat{G}(T_i)^{-2} \, \mathbf{1}(T_i < T_j,\ T_i < \tau)\, \mathbf{1}(\eta_i > \eta_j)}{\displaystyle\sum_i \sum_j \Delta_i \, \hat{G}(T_i)^{-2} \, \mathbf{1}(T_i < T_j,\ T_i < \tau)}. \tag{2}$$

In plain language: a death at a time when few patients remain uncensored is rare to observe, so it counts more. Only the ordering of the scores η_i matters. The challenge computes C_τ with `scikit-survival` [4].

4 Data audit (Stage 1)

4.1 Linking and outcome

All IDs are unique, every mutation row belongs to a known patient, and no patient appears in both training and test sets. However, 150 training patients have no recorded outcome and are excluded from modelling. Table 2 summarises the remaining outcomes and Figure 1 shows the observed times.

Table 2: Outcome of the training patients.

Quantity	Value
Training patients	3,323
Without outcome (excluded)	150
With outcome	3,173
Deaths ($\Delta = 1$)	1,600 (50.4%)
Censored ($\Delta = 0$)	1,573 (49.6%)
Median observed time (years)	1.65
Maximum observed time (years)	22.04
Observed time exactly 0	53
Observed beyond the 7-year horizon	172

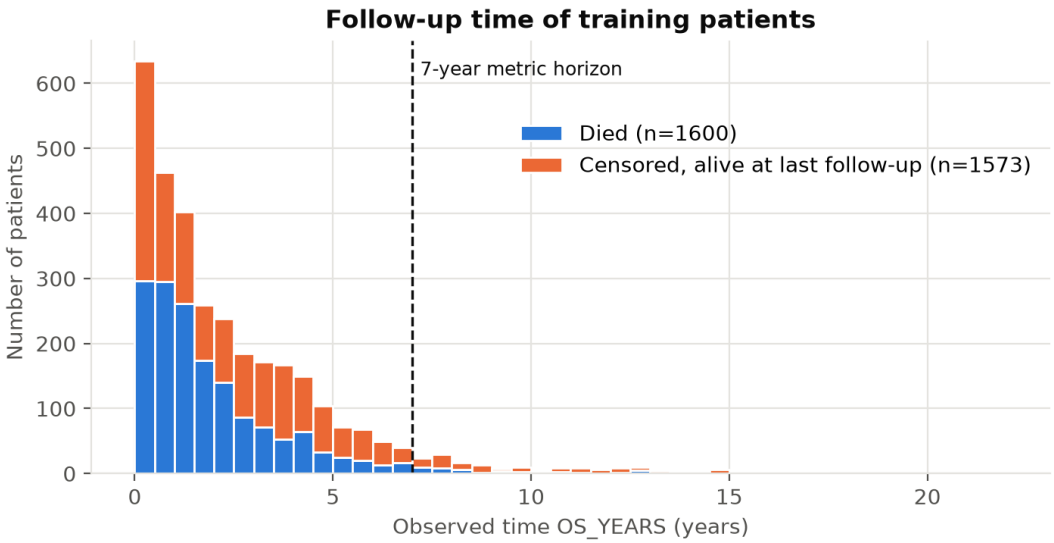


Figure 1: Observed time of the 3,173 training patients with an outcome, split into deaths and censored patients. The dashed line marks the 7-year horizon of the metric.

About half the patients are censored, so simply discarding them or treating their observed time as a death time would badly bias any model. Most deaths occur within the first two years, and only 172 patients are followed beyond seven years, which justifies the truncation of the metric.

4.2 Clinical centres: the test set is a new hospital

The 3,323 training patients come from 23 centres, while all 1,193 test patients come from one centre, `KYW`, that does not appear in training (Figure 2). Death rates also vary widely between

training centres, from 11% (ROM) to 92% (MUV), reflecting different patient populations and follow-up lengths.

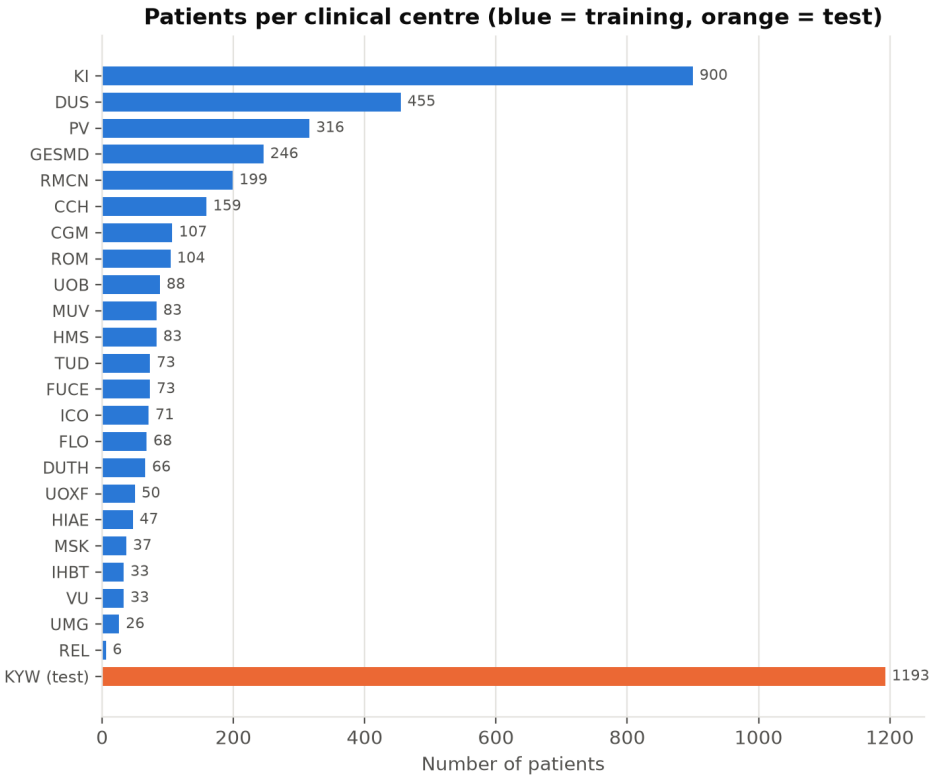


Figure 2: Number of patients per clinical centre. The test centre KYW (orange) contributes no training patients.

This has a direct consequence for validation. A random K-fold split mixes patients from the same centre into training and validation folds, so the model is always validated on centres it has already seen. The real task is to predict for an *unseen* centre. Cross-validation will therefore be grouped by centre, and the CENTER variable cannot be used as a predictor.

4.3 Clinical variables: training versus test

Table 3 and Figures 3 and 4 compare the clinical variables in the two sets.

Table 3: Missing values and medians of the clinical variables.

	Missing (%)		Median	
Variable	Train	Test	Train	Test
BM_BLAST (%)	3.3	9.6	3.00	6.00
WBC (G/L)	8.2	9.4	4.10	2.90
ANC (G/L)	5.8	11.8	2.00	1.30
MONOCYTES (G/L)	18.1	74.0	0.37	0.20
HB (g/dL)	3.3	9.3	9.70	8.40
PLT (G/L)	3.7	9.6	123.00	74.00
CYTOGENETICS	11.6	9.7	–	–

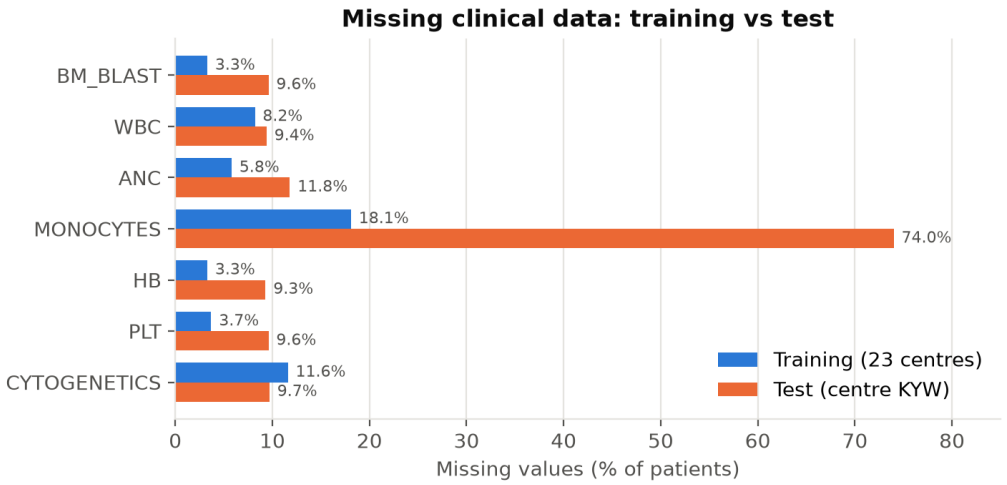


Figure 3: Percentage of missing values in each clinical variable.

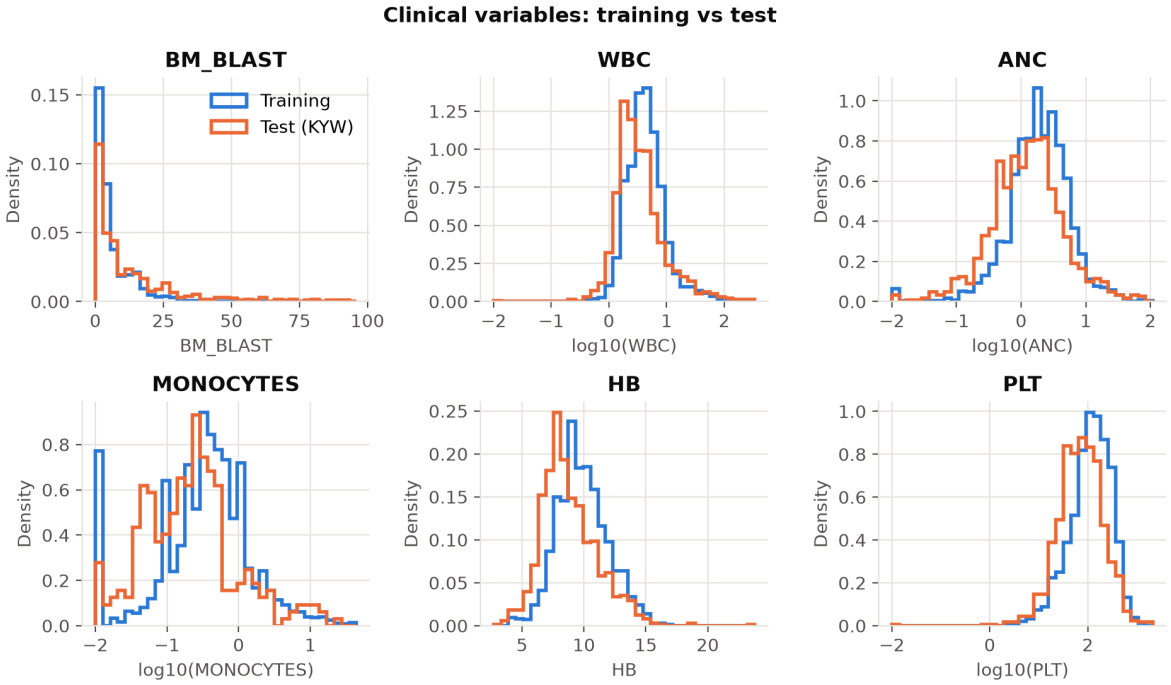


Figure 4: Distributions of the clinical variables in training (blue) and test (orange). Each histogram is scaled to area one so that sets of different sizes can be compared. Skewed blood counts are shown on a $\log_{10}$ scale; the spike at -2 for monocytes corresponds to recorded values of zero.

The test patients are clinically sicker: lower haemoglobin and platelets, lower neutrophils, and a much larger share with bone-marrow blasts above 20% (17.6% versus 4.2%). Monocytes are missing for 74% of test patients, so a model that leans on that variable would be fragile on the test set.

4.4 Gene mutations

Training patients carry on average 3.29 recorded mutations and test patients 2.59; 297 training and 139 test patients have none, which is genuine information rather than missing data. Training

data contain 124 distinct genes and test data 83, of which 19 never appear in training. Of the training rows, 114 lack a chromosomal position: they are partial tandem duplications of *MLL* (89) and internal tandem duplications of *FLT3* (25), which are structural events rather than single-base changes. In 1,274 cases the same gene is mutated more than once in one patient.

Table 4: Ten most frequently mutated genes in training: percentage of patients with at least one mutation.

Gene	Train (%)	Test (%)
TET2	31.1	17.5
ASXL1	26.9	12.7
SF3B1	22.2	7.7
SRSF2	17.2	9.1
DNMT3A	16.1	11.1
RUNX1	13.9	13.7
TP53	11.4	25.7
STAG2	9.1	6.7
U2AF1	8.6	8.4
EZH2	6.5	6.7

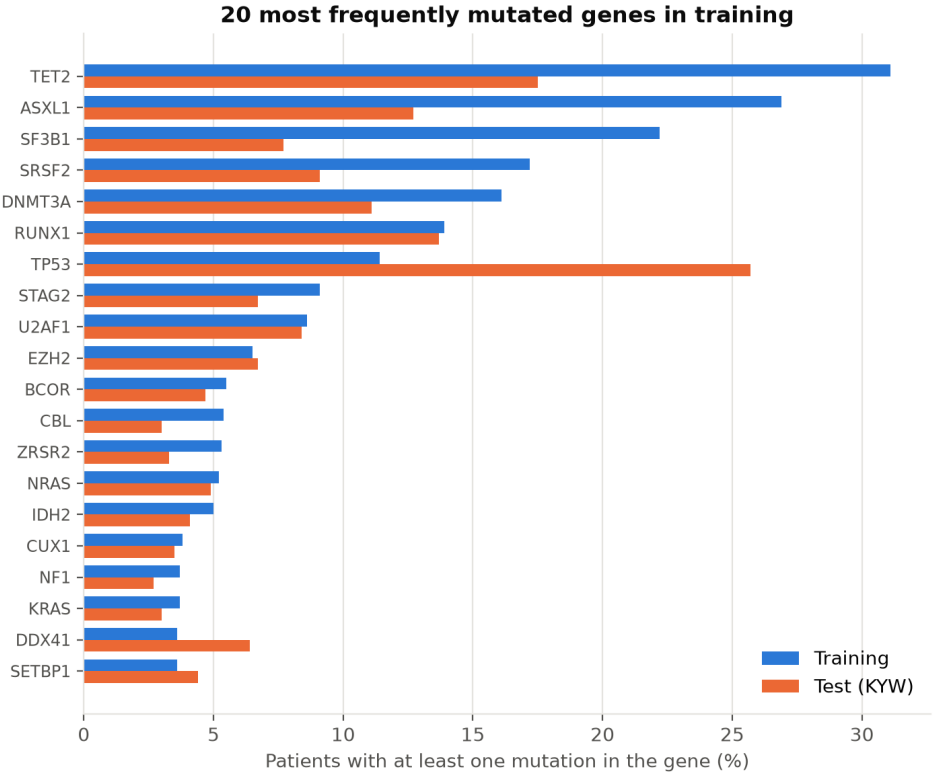


Figure 5: The twenty most frequently mutated genes in training, with their frequency in the test centre.

The most striking difference is *TP53*, mutated in 25.7% of test patients against 11.4% in training (Table 4, Figure 5). *TP53* mutations are associated with poor prognosis in myeloid neoplasms, which is consistent with the sicker clinical profile of the test centre. By contrast, *SF3B1* mutations, typically associated with lower-risk disease, are almost three times rarer in

the test set.

4.5 Cytogenetics text

In training, karyotypes are written in lower case (`46,xy`); in the test set they are written in upper case (`46,XY`), and 241 test patients have the single word `Normal`. Any parsing must first convert text to lower case and map `Normal` to a normal karyotype; otherwise every test karyotype would look unfamiliar to the model.

4.6 Consequences for modelling

1. Exclude the 150 training patients without outcome.
2. Validate with cross-validation grouped by centre, and do not use `CENTER` as a predictor.
3. Prefer features that are measured consistently in both sets; treat `MONOCYTES` with caution.
4. Normalise the cytogenetics text before extracting features.
5. Encode "no mutation found" as zero, not as missing.

5 Next steps

Stage 2 will estimate Kaplan–Meier survival curves overall and by clinical and genetic groups, compare groups with log-rank tests, and compute the C-index and the IPCW C-index by hand on a small example before checking them against `scikit-survival`.

References

[1] D. P. Noren *et al.*, A crowdsourcing approach to developing and assessing prediction algorithms for AML prognosis, *PLOS Computational Biology*, 12(6): e1004890, 2016.

[2] F. E. Harrell, R. M. Califf, D. B. Pryor, K. L. Lee and R. A. Rosati, Evaluating the yield of medical tests, *JAMA*, 247(18): 2543–2546, 1982.

[3] H. Uno, T. Cai, M. J. Pencina, R. B. D'Agostino and L. J. Wei, On the C-statistics for evaluating overall adequacy of risk prediction procedures with censored survival data, *Statistics in Medicine*, 30(10): 1105–1117, 2011.

[4] S. Pölsterl, scikit-survival: a library for time-to-event analysis built on top of scikit-learn, *Journal of Machine Learning Research*, 21(212): 1–6, 2020.